Susceptible–Infected–Recovered (SIR) Model: A Simple Framework to Study COVID-19 Spread

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Abstract:

The COVID-19 pandemic highlighted the importance of quantitative modelling for analysing infectious disease dynamics. This study applies the Susceptible-Infected-Recovered (SIR) model (Kermack & McKendrick, 1927) to investigate COVID-19 transmission in India using model-generated epidemic trajectories. Daily estimates of susceptible, infected, and recovered populations from 29 January 2020 to 29 January 2021 were obtained from a publicly available Kaggle dataset derived from epidemiological modelling, chosen for its internal consistency with the SIR framework rather than as a source of empirical surveillance data. The SIR framework was implemented numerically, and finite-difference approximations were used to estimate time-varying transmission and recovery rates. To ensure numerical stability, parameter estimation was restricted to a stable phase of the epidemic with sufficiently large infection counts. Mean parameter estimates yielded a large model-implied basic reproduction number, reflecting strong transmission dynamics within the assumed framework. Temporal stability of the estimated parameters supported the use of average rates to summarise epidemic dynamics. Graphical analysis of epidemic curves and parameter trajectories illustrated key features of the model-generated outbreak dynamics. While the simplifying assumptions of the SIR model limit its realism for COVID-19, the analysis demonstrates the model's value for parameter estimation, trend analysis, and pedagogical insight in epidemic modelling.

Keywords:

SIR Model, COVID-19 epidemiology, Infectious Disease Dynamics, Parameter Estimation, Graphical Interpretation.

1. Introduction

The global COVID-19 pandemic, caused by a severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), had an unprecedented impact on global health, economies, and social life. As countries faced a rapid increase in case numbers and limited medical resources, understanding and controlling the spread of infectious diseases became essential. Mathematical and statistical models emerged as essential tools for analysing and predicting epidemic dynamics. Among these, epidemiological models, particularly the Susceptible-Infected-Recovered (SIR) model introduced by Kermack & McKendrick (1927), remains a foundational tool in epidemiological theory (Hethcote, 2000) for studying infectious disease transmission dynamics.

2. Epidemiological Modelling in Understanding Pandemics

Epidemiological models are simplified mathematical representations of contagious disease outbreaks, forming the theoretical basis of modern infectious-disease modelling (Hethcote, 2000). They translate real-world interactions into structured equations to provide insights into disease dynamics. They played an important role in informing public health measures to manage the COVID-19 pandemic and in determining which interventions were most likely to reduce case numbers.

The SIR framework is a classic epidemiological model that represents disease transmission by using parameters for infection and recovery, enabling analysis of outbreak growth, peak infection periods, and eventual decline. Rather than establishing real-world causation, compartmental models such as SIR are best understood as conceptual approximations.

3. The Susceptible-Infected-Recovered (SIR) Model

The Susceptible-Infected-Recovered (SIR) Model is a deterministic compartmental modelling framework used to describe the spread of infectious diseases, originally formulated by Kermack and McKendrick (1927).

At any time t during the COVID-19 outbreak, the total population N is divided into three mutually exclusive compartments:

- $S(t)$: number of susceptible individuals who can contract the disease
- $I(t)$: number of infected individuals who can transmit the disease
- $R(t)$: number of individuals who have recovered with permanent immunity

The SIR model describes a one-directional flow of individuals between the compartments. The model assumes a fixed population size and is given by:

$$N = S(t) + I(t) + R(t)$$

I. Conceptual Framework and Assumptions

The SIR model relies on several assumptions that enable mathematical tractability but limit direct real-world applicability.

i. Closed Population: The total population size is assumed to remain constant over the period of analysis, with no births, deaths, immigration, or emigration.
ii. Homogeneous Mixing: Every individual in the population is assumed to have an equal probability of encountering any other individual in the population.
iii. Instantaneous Transition between Compartments: Individuals move directly from the susceptible to infected compartment upon infection and from infected to recovered upon recovery.
iv. Permanent Immunity after Recovery: Recovered individuals are assumed to gain permanent immunity within the model structure.
v. Constant Transmission and Recovery Rates: The transmission rate (β) and recovery rate (γ) are assumed to remain constant throughout the study period.
vi. No Vaccination Effects: The model assumes an entirely unvaccinated population.

Further, COVID-19 related deaths and asymptomatic infections are not modelled separately and are implicitly included in the recovered compartment.

II. Mathematical Foundation

Let $S(t)$, $I(t)$, and $R(t)$ denote the number of susceptible, infected, and recovered individuals at time t during the COVID-19 outbreak. The dynamics of the SIR model are described by the following system of ordinary differential equations:

$$\frac{dS}{dt} = -\frac{\beta\, S(t)I(t)}{N}$$

$$\frac{dI}{dt} = \frac{\beta\, S(t)I(t)}{N} - \gamma I(t)$$

$$\frac{dR}{dt} = \gamma I(t)$$

The parameter β denotes the transmission rate, while γ denotes the recovery rate.
These equations describe how individuals move between compartments over time and form the basis for analysing COVID-19 epidemic dynamics within the SIR modelling framework.

III. Model Parameters and Their Interpretation:

The SIR model is characterised by two key parameters that determine the transmission and progression of the disease:

a) Transmission Rate: The transmission rate (β) measures how quickly susceptible individuals become infected through contact with infectious individuals. It reflects both contact frequency and transmission probability and plays a key role in capturing the spread of COVID-19.
b) Recovery Rate: The recovery rate (γ) represents the rate at which infected individuals recover from the infectious population. The average infectious period is given by $\frac{1}{\gamma}$, indicating the expected duration for which an individual remains infectious.

The basic reproduction number R_0 is defined as $R_0 = \frac{\beta}{\gamma}$ (Kermack & McKendrick, 1927; Hethcote, 2000).
R_0 represents the average number of secondary infections predicted by the model for a single infected individual in a fully susceptible population.
When $R_0 > 1$, the model predicts epidemic growth, while $R_0 < 1$ indicates eventual decline, making R_0 a critical indicator of COVID-19 transmission intensity.

IV. Dataset Description

The analysis employs India's COVID-19 SIR estimates dataset, obtained from Kaggle, covering the period from 29 January 2020 to 29 January 2021. This dataset provides daily model-generated secondary estimates of the susceptible $S(t)$, infected $I(t)$, and recovered $R(t)$ populations, derived from epidemiological modelling, and was used because its internally consistent SIR trajectories permit methodological exploration and parameter estimation within the same modelling framework. The data are organised by date, with each entry representing model-predicted transmission dynamics for that day. These daily estimates allow detailed tracking of the model-generated epidemic progression over time, capturing key trends in disease spread across India. As these are secondary model-derived estimates rather than raw

surveillance data, the analysis does not attempt to recover true epidemic parameters, and findings should not be interpreted as empirical measurements of actual COVID-19 transmission in India.

V. Model Implementation and Results

i. Model Implementation

The SIR model was implemented numerically to generate time-series trajectories of the susceptible, infected, and recovered populations, which were treated as internally consistent, model-generated epidemic trajectories and used for parameter estimation.

Since the model is continuous in time, finite-difference approximations were used to estimate daily values of the transmission and recovery rates from the simulated data:

$$\beta(t) \approx -\frac{N}{S(t)I(t)}\frac{\Delta S(t)}{\Delta t}$$

$$\gamma(t) \approx \frac{1}{I(t)}\frac{\Delta R(t)}{\Delta t}$$

where $\Delta t = 1$ day.

ii. Parameter Estimates and Interpretation

To avoid unstable estimates caused by very small numbers of infected individuals, parameter estimation was restricted to a stable phase of the epidemic, beginning on day 143, when the infected population exceeded 1% of its maximum value.
Daily estimates of $\beta(t)$ and $\gamma(t)$ were then averaged over this period to obtain mean parameter estimates that summarise the overall epidemic dynamics implied by the model.

The estimated transmission rate $\hat{\beta} = 0.184$ indicated a high effective contact rate within the SIR framework, meaning that interactions between susceptible and infected individuals frequently led to new infections. The estimated recovery rate $\hat{\gamma} = 0.0196$ implied an average infectious period of approximately $\frac{1}{\hat{\gamma}} \approx 51$ days, indicating relatively slow recovery of infected individuals with permanent immunity.

Together, these estimates yielded a basic reproduction number $R_0 = 9.42$, suggesting that, in a fully susceptible population, each infected individual would generate more than nine

secondary infections on average. Such a high R_0 reflects extremely strong transmission dynamics implied by the model assumptions.

Day-wise estimates of β and γ showed strong temporal stability. The transmission rate varied between 0.176 and 0.204, while the recovery rate remained tightly clustered around 0.0195, supporting the use of mean parameter values to summarise epidemic behaviour.

VI. Diagrammatic Representation of the Model

a. SIR Epidemic Curves Over Time

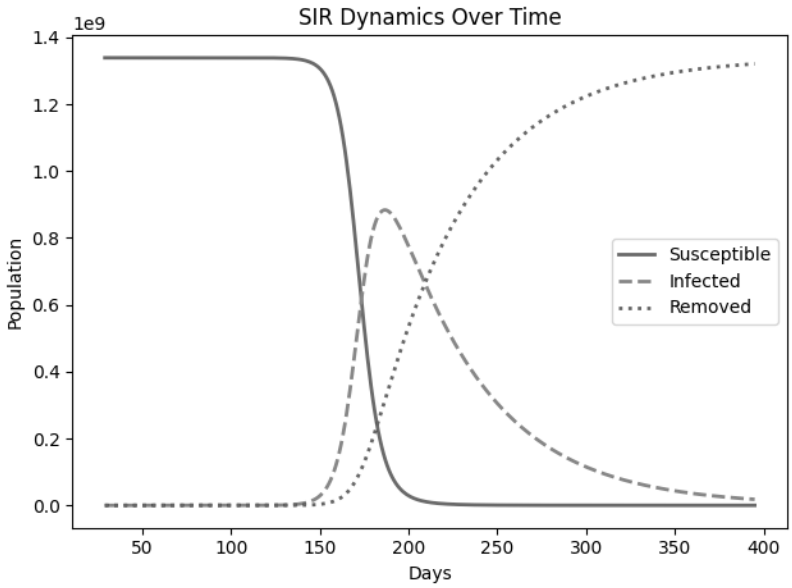


Figure 1: Temporal Evolution of Susceptible, Infected, and Removed Populations

Figure 1 illustrates the internally consistent epidemic behaviour implied by the SIR framework, highlighting susceptible depletion, a single infection peak, and cumulative recoveries. The susceptible population remains nearly constant initially, then declines sharply as infections rise. The infected curve exhibits a single pronounced peak, indicating the point of maximum disease spread, after which infections decline as susceptible individuals become scarce. Over time, the recovered population increases steadily and eventually dominates, reflecting cumulative recoveries over time.

b. Infected Population Dynamics During the Epidemic

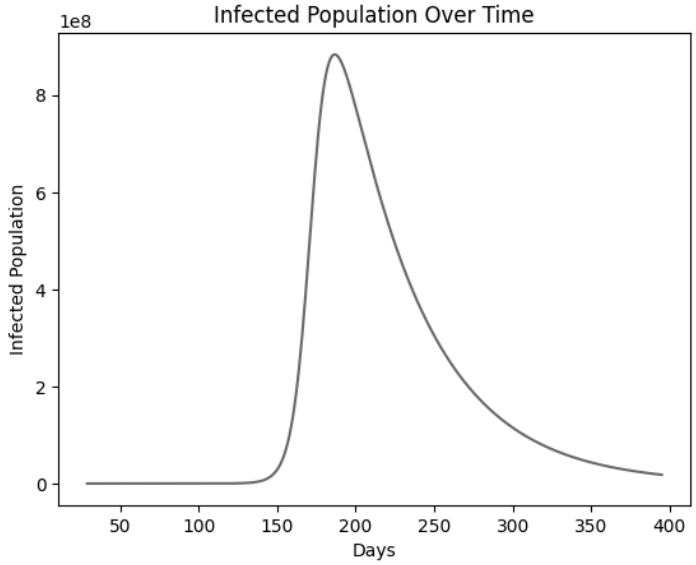


Figure 2: Infected Population Over Time

Figure 2 highlights the rise and decline of the infected population driven by susceptible depletion and recovery dynamics within the model.

c. Temporal Variation of Daily Transmission Rate

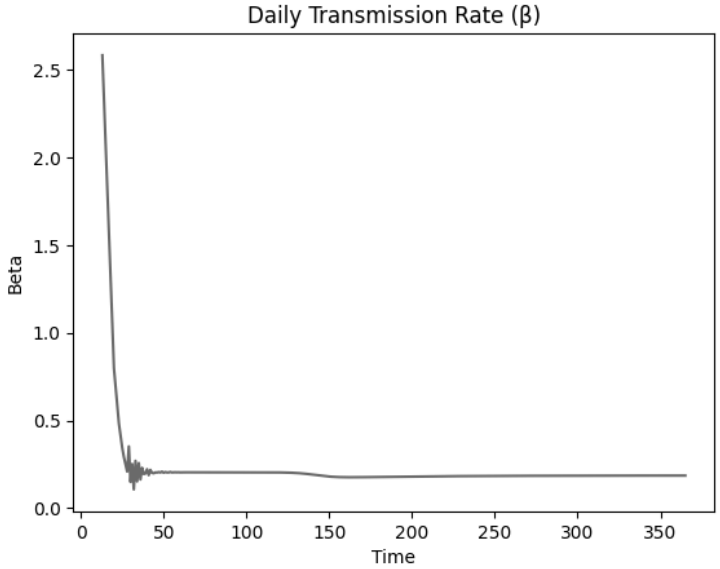


Figure 3: Daily Transmission Rate Over Time

Figure 3 demonstrates that initial volatility in the estimated transmission rate arises from numerical sensitivity during early low-infection periods, followed by stabilisation that supports the use of an average transmission rate.

VII. Limitations of the SIR Model

While the SIR model provides a simple framework for studying infectious disease spread, its application to COVID-19 has important limitations. The framework does not include an explicit incubation or exposed stage, which is significant in the transmission of SARS-CoV-2. It does not account for immunity gained through vaccination or its impact on transmission dynamics. It further assumes homogeneous population mixing and constant transmission and recovery rates, whereas real-world transmission is strongly influenced by behavioural changes, such as lockdowns, public health interventions, mask mandates, vaccination, and the emergence of new variants. The assumption of permanent immunity is also questionable due to documented cases of reinfections and waning immunity. An additional limitation arises from the reliance on secondary, model-generated SIR estimates, which inherit the assumptions and structural constraints of the original modelling process and may amplify simplifications when re-analysed. Consequently, the SIR model is best suited for understanding general epidemic trends rather than making precise real-world COVID-19 predictions.

4. Policy Implications and Public Health Relevance

The SIR model provides valuable qualitative insights for public health planning. By identifying peak infection periods and epidemic duration, it helps policymakers anticipate healthcare demand and allocate resources effectively. Variations in transmission rates can be interpreted as model-consistent representations of changes in transmission intensity, without implying direct causation.

5. Conclusion

The Susceptible–Infected–Recovered model remains a foundational framework for understanding infectious disease dynamics. While its assumptions and reliance on secondary model-generated data limit its direct applicability to real-world COVID-19 dynamics, the model provides clear insights into parameter relationships and epidemic trajectories within a

simplified framework. As such, it serves as an important foundation for developing more complex and realistic epidemiological models.

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